

On Multi-Marginal Partial Optimal Transport: Rectifying Infeasible Extension Strategies and Efficient Primal-Dual Methods

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Editor: My editor

Abstract

We study the Multi-marginal Partial Optimal Transport (MMPOT)—a generalization of the discrete Optimal Transport (OT) problem—over m unbalanced marginals, each with at most n support points. Previous works in the POT/MMPOT literature utilize an extension strategy to reduce the problem to the OT formulation, and then leverage prominent computational Entropic OT methods to solve it. In this work, we show that such strategy not only produces infeasible solutions for the entropic regularized MMPOT problem, but also provably worsens the performance of these methods compared to its OT counterparts. To this end, we take a primal-dual approach and obtain competitive convergence guarantees for several methods, namely PDAAM, APDAGD and GreenkhornMMPOT, which provides a complete picture for the computational POT literature. Extensive numerical experiments at various scales showcase the effectiveness of the proposed algorithms.

1 Notations

Let $[n] := \{1, 2, \dots, n\}$ and $\mathbb{R}_+^n := \{u \in \mathbb{R}^n : u_i \geq 0, \forall i \in [n]\}$. Let $1_n \in \mathbb{R}^n$ denote the all-ones vector and define the probability simplex

$$\Delta_n := \{u \in \mathbb{R}_+^n : 1_n^\top u = 1\}.$$

For a set S , $|S|$ denotes its cardinality.

For a differentiable function f , ∇f denotes its gradient, and $\nabla_\beta f$ the gradient with respect to the variable β . For $x \in \mathbb{R}^n$, $\|x\|_p$ denotes the ℓ_p -norm ($1 \leq p \leq \infty$), and $\|x\| := \|x\|_2$. For vectors x, y of equal dimension, $\min\{x, y\}$ denotes the component-wise minimum. Let $A \in \mathbb{R}^{n_1 \times \dots \times n_m}$ be an m -way tensor with entries $A(i_1, \dots, i_m)$. We define

$$\|A\|_\infty := \max_{i_k \in [n_k], k \in [m]} |A(i_1, \dots, i_m)|, \quad \|A\|_1 := \sum_{i_k \in [n_k], k \in [m]} |A(i_1, \dots, i_m)|.$$

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For each $j \in [m]$, the j -th marginal operator $r_j : \mathbb{R}^{n_1 \times \dots \times n_m} \rightarrow \mathbb{R}^{n_j}$ is defined by

$$[r_j(A)]_{i_j} := \sum_{\substack{i_1, \dots, i_{j-1} \\ i_{j+1}, \dots, i_m}} A(i_1, \dots, i_m).$$

For tensors $A, B \in \mathbb{R}^{n_1 \times \dots \times n_m}$, their Frobenius inner product is

$$\langle A, B \rangle := \sum_{i_k \in [n_k], k \in [m]} A(i_1, \dots, i_m) B(i_1, \dots, i_m).$$

Given the dimension n and accuracy ε , the notation $a = \mathcal{O}(b(n, \varepsilon))$ means that $a \leq C b(n, \varepsilon)$ for some constant $C > 0$ independent of n and ε . The notation $a = \tilde{\mathcal{O}}(b(n, \varepsilon))$ allows additional logarithmic factors:

$$a \leq C b(n, \varepsilon) \log^k(n, \varepsilon)$$

for some constants $C > 0$ and $k > 0$ independent of n and ε . Let $X \in \mathbb{R}^{n \times \dots \times n}$ be an m -way tensor and $p_j \in \mathbb{R}_+^n$ for $j \in [m]$. We define

$$X \in \Pi(p_1, \dots, p_m) \iff r_j(X) = p_j, \quad \forall j \in [m].$$

For a total mass $s \geq 0$ and slack vectors $q_j \in \mathbb{R}_+^n$, we define

$$X \in \Pi_s(p_1, \dots, p_m) \iff \begin{cases} r_j(X) + q_j = p_j, & \forall j \in [m], \\ \langle X, 1_{n_1 \times n_2 \times \dots \times n_m} \rangle = s, \\ X \geq 0, \quad q_j \geq 0. \end{cases}$$

2 Preliminaries

In this section, we first formulate MMPOT as a Linear Programming (LP) problem with additional slack variables that account for the inequality constraints. To obtain an approximate solution, we introduce an entropy regularizer to this LP and adopt a primal-dual setting. By showing that the resulting objective function is strongly convex on a simple closed convex set, the dual problem can be derived as a smooth unconstrained convex optimization problem. Last but not least, we establish several useful properties of this dual, including a tight upper bound on the optimal solution and estimates of the Lipschitz smoothness constants with respect to the ℓ_2 and ℓ_∞ norms.

2.1 Asymmetric smooth convex dual derivation

Consider m discrete measures with n support points each. We denote these measures as p_i , $\forall i \in [m]$. Given the cost tensor $C \in \mathbb{R}^{n \times \dots \times n}$, the MMPOT problem can be described as the following linear programming problem:

$$\min_{X \in \Pi_s(p_1, \dots, p_m)} \langle C, X \rangle. \tag{1}$$

While one can define the constraint in Problem (1) as an inequality instead of using slack variables q_j since the objective doesn't rely on such variables, later on we will show that our formulation is especially useful in the primal-dual setting, where the dual derivation construction shows the difference between OT and POT clearly. Next, we define an ε -approximate multimarginal transportation plan as follows.

Definition 1. A nonnegative tensor $X_b \in \mathbb{R}_+^{n \times \dots \times n}$ is called an ε -approximate multimarginal transportation plan if there exist slack variables q_i such that the constraint $X_b \in \Pi_s(p_1, \dots, p_m)$ is satisfied and

$$\langle C, X_b \rangle \leq \langle C, X^* \rangle + \varepsilon, \quad (2)$$

where X^* is the optimal multimarginal transportation plan of the MMPO problem.

Next, we simplify formulation (1) and derive an equivalent linear-constrained convex optimization problem as follows. Let

$$x := (\text{vec}(X)^\top, q_1^\top, q_2^\top, \dots, q_m^\top)^\top, \quad d := (\text{vec}(C)^\top, 0_{mn}^\top)^\top. \quad (3)$$

We construct a linear transformation A such that

$$(Ax)^\top = ((r_1(X) + q_1)^\top, (r_2(X) + q_2)^\top, \dots, \langle X, 1 \rangle), \quad (4)$$

and

$$b^\top = (p_1^\top, p_2^\top, \dots, p_m^\top, s). \quad (5)$$

Hence, the constraints of the above formulation can be encoded as $Ax = b$, yielding

$$\min_{x \geq 0} f(x) = \langle d, x \rangle \quad \text{s.t.} \quad Ax = b. \quad (6)$$

Next, we define the following simple closed convex set

$$Q := \left\{ x = (a^\top, b^\top)^\top : a \in \mathbb{R}^{n^m}, b \in \mathbb{R}^{mn}, x \geq 0, \|a\|_1 = s, \|b\|_1 = \sum_{i=1}^m \|p_i\|_1 - ms \right\}. \quad (7)$$

Intuitively, the defined convex set contains all solutions that satisfy the constraints in Problem 1. Before proceeding, we prove that on Q , the function $f(x)$ is strongly convex with respect to l_1 -norm.

Lemma 2. Let

$$D_i := \|p_i\|_1 - s, \quad i = 1, \dots, m, \quad (8)$$

and define

$$Q := \left\{ x = (a^\top, b_1^\top, \dots, b_m^\top)^\top : a \in \mathbb{R}^{n^m}, b_i \in \mathbb{R}^n, \right. \\ \left. x \geq 0, \|x\|_1 = s + \sum_{i=1}^m D_i \right\}. \quad (9)$$

Then the entropic regularized objective

$$f(x) = \sum_{i=1}^{n^m+mn} (\gamma x_i \log x_i + d_i x_i) \quad (10)$$

is $\frac{\gamma}{s+\sum_{i=1}^m D_i}$ -strongly convex with respect to the ℓ_1 -norm over Q .

Proof A function f is $\frac{\gamma}{s+\sum_{i=1}^m D_i}$ -strongly convex with respect to the ℓ_1 -norm on Q if, for all $x, y \in Q$,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{\gamma}{s + \sum_{i=1}^m D_i} \|x - y\|_1^2. \quad (11)$$

For $f(x) = \sum_i (\gamma x_i \log x_i + d_i x_i)$, we have

$$\frac{\partial f(x)}{\partial x_i} = \gamma \log x_i + \gamma + d_i. \quad (12)$$

Hence the above condition is equivalent to

$$\sum_{i=1}^{n^m+mn} \gamma (x_i - y_i) (\log x_i - \log y_i) \geq \frac{\gamma}{s + \sum_{i=1}^m D_i} \|x - y\|_1^2. \quad (13)$$

Dividing both sides by $\gamma > 0$, it suffices to show

$$\sum_{i=1}^{n^m+mn} (x_i - y_i) (\log x_i - \log y_i) \geq \frac{1}{s + \sum_{i=1}^m D_i} \|x - y\|_1^2. \quad (\star)$$

For $\lambda \in [0, 1]$, define

$$s_1(\lambda) := \sum_{i=1}^{n^m+mn} (x_i - y_i) \log(y_i + \lambda(x_i - y_i)), \quad (14)$$

$$s_2(\lambda) := \lambda \|x - y\|_1^2. \quad (15)$$

Their derivatives are

$$s'_1(\lambda) = \sum_{i=1}^{n^m+mn} \frac{(x_i - y_i)^2}{y_i + \lambda(x_i - y_i)}, \quad (16)$$

$$s'_2(\lambda) = \|x - y\|_1^2. \quad (17)$$

Since $x, y \in Q$ and $\|x\|_1 = \|y\|_1 = s + \sum_{i=1}^m D_i$, we have

$$\sum_{i=1}^{n^m+mn} (y_i + \lambda(x_i - y_i)) = s + \sum_{i=1}^m D_i, \quad \forall \lambda \in [0, 1]. \quad (18)$$

Applying the Cauchy–Schwarz inequality yields

$$s'_1(\lambda) \geq \frac{(\sum_{i=1}^{n^m+mn} |x_i - y_i|)^2}{\sum_{i=1}^{n^m+mn} (y_i + \lambda(x_i - y_i))} = \frac{\|x - y\|_1^2}{s + \sum_{i=1}^m D_i}. \quad (19)$$

Therefore,

$$(s + \sum_{i=1}^m D_i) s'_1(\lambda) \geq s'_2(\lambda), \quad \forall \lambda \in [0, 1]. \quad (20)$$

By Cauchy's mean value theorem, there exists $c \in (0, 1)$ such that

$$(s + \sum_{i=1}^m D_i) (s_1(1) - s_1(0)) \geq s_2(1) - s_2(0). \quad (21)$$

Substituting

$$s_1(1) - s_1(0) = \sum_i (x_i - y_i)(\log x_i - \log y_i), \quad (22)$$

$$s_2(1) - s_2(0) = \|x - y\|_1^2, \quad (23)$$

we obtain

$$\sum_{i=1}^{n^m+mn} (x_i - y_i)(\log x_i - \log y_i) \geq \frac{1}{s + \sum_{i=1}^m D_i} \|x - y\|_1^2, \quad (24)$$

which proves $(\star)$ and completes the proof. ■

To this end, by abuse of notation of $x = (\text{vec}(X)^\top, q_1^\top, q_2^\top, \dots, q_m^\top)^\top$ define the Lagrangian function as:

$$\begin{aligned} \mathcal{L}(x, a_1, \dots, a_m, s_1) &= \langle C, X \rangle + \eta \langle X, \log X \rangle + \sum_{j=1}^m \eta \langle q_j, \log q_j \rangle \\ &\quad + \sum_{i=1}^m \langle a_i, r_i(X) + q_i - p_i \rangle + s_1 (\langle X, 1_{n^m} \rangle - s). \end{aligned} \quad (25)$$

Since all value of X and q_i that satisfies $X \in \Pi_s(p_1, \dots, p_m)$ implies $x \in Q$, we consider the following minimization problem to derive the dual:

$$\min_{x \in Q \cap X \in \Pi_s(p_1, \dots, p_m)} \mathcal{L}(x, a_1, a_2, \dots, a_m, s). \quad (26)$$

Finally, let $D = s + \sum_{i=1}^m D_i$, simple computation yields:

$$X_{i_1, \dots, i_m} = \frac{D \exp\left(-\frac{1}{\eta} \left(\sum_{t=1}^m a_{t, i_t} + C_{i_1, \dots, i_m} + s_1\right)\right)}{\sum_{j_1, \dots, j_m=1}^n \exp\left(-\frac{1}{\eta} \left(\sum_{t=1}^m a_{t, j_t} + C_{j_1, \dots, j_m} + s_1\right)\right) + \sum_{k=1}^m \left\| \exp\left(-\frac{1}{\eta} a_k\right) \right\|_1}. \quad (27)$$

$$q_i = \frac{D \exp\left(-\frac{1}{\eta} a_i\right)}{\sum_{j_1, \dots, j_m=1}^n \exp\left(-\frac{1}{\eta} \left(\sum_{t=1}^m a_{t, j_t} + C_{j_1, \dots, j_m} + s_1\right)\right) + \sum_{k=1}^m \left\| \exp\left(-\frac{1}{\eta} a_k\right) \right\|_1}. \quad (28)$$

Plug in these values to the Lagrangian, we get the following problem:

$$\begin{aligned} \max_{a_1, \dots, a_m} \quad & \Phi(a_1, \dots, a_m, s_1) = -\langle s_1, s \rangle - \sum_{i=1}^m \langle a_i, p_i \rangle \\ & - \eta D \log \left(\sum_{i_1, \dots, i_m} \exp\left(-\frac{\sum_{t=1}^m a_{t, i_t} + C_{i_1, \dots, i_m} + s_1}{\eta}\right) + \sum_{k=1}^m \left\| \exp\left(-\frac{1}{\eta} a_k\right) \right\|_1 \right). \end{aligned} \quad (29)$$

Let $u_i = -\eta^{-1} a_i$ and $t = -\eta^{-1} s_1$, we arrive at the following minimization problem, which we call *the first dual entropic MMOT problem*

$$\begin{aligned} \min_{u_1, \dots, u_m, t} \quad & \phi(u_1, \dots, u_m, t) := D \log \left(\sum_{i_1, \dots, i_m} \exp\left(t + \sum_{k=1}^m u_{k, i_k} - \frac{1}{\eta} C_{i_1, \dots, i_m}\right) + \sum_{k=1}^m \left\| \exp(u_k) \right\|_1 \right) \\ & - \sum_{k=1}^m \langle u_k, p_k \rangle - \langle t, s \rangle. \end{aligned} \quad (30)$$

where we have use the notation $u_{i,j} = u_i(j)$ to denote the j -th entry of the vector u_i . One can observe that, for the special case of $\|q_i\|_1 = s \ \forall i \in [1, m]$, we must have $D = s$, thus our dual degenerated into the *dual for entropic MMOT* as derived in Lin et al. (2022) since the additional "sum-of-exponential" $\sum_{k=1}^m \left\| \exp(u_k) \right\|_1 = 0$, and the need for t is eliminated by taking the $\exp(t)$ term out.

2.1.1 UPPER BOUND ON ℓ_∞ OF SOLUTION

Next, we show that there exists a certain solution $u^* = (u_1^{*\top}, \dots, u_m^{*\top})^\top$ such that its ℓ_∞ norm scales asymptotically as $\mathcal{O}(\frac{\|C\|_\infty}{\eta})$. This in turn implies that a rough bound on the ℓ_2 of this solution is asymptotically $\mathcal{O}(\frac{\|C\|_\infty \sqrt{nm}}{\eta})$. Combine with the fact that the global Lipschitz smoothness constant with respect to ℓ_2 norm of Problem (30) is asymptotically $\mathcal{O}(\sqrt{m})$, one can readily obtain computational complexity of classical gradient descent based algorithm.

Lemma 3. *For the dual entropic regularized multi-marginal partial optimal transport problem (30), assume that*

$$p_{k,j} > 0, \quad k \in [m], \quad j \in [n],$$

and

$$0 < s < \min_{k \in [m]} \|p_k\|_1.$$

Let

$$d_k := \|p_k\|_1 - s, \quad d_{\min} := \min_{k \in [m]} d_k, \quad d_{\max} := \max_{k \in [m]} d_k,$$

and define

$$\rho := \log \frac{d_{\max}}{d_{\min}}.$$

Furthermore, let

$$\mathcal{R} := \frac{\|C\|_\infty}{\eta} + \log \left(\frac{\max_{k \in [m]} \|p_k\|_1}{\min_{k \in [m], j \in [n]} p_{k,j}} \right).$$

Then there exists an optimal solution

$$u^* = (u_1^{*\top}, \dots, u_m^{*\top})^\top, \quad t^*,$$

such that

$$\|u^*\|_\infty \leq \mathcal{R} + \frac{\rho}{2}.$$

Moreover, for fixed marginals and fixed transported mass s ,

$$|t^*| = \mathcal{O} \left(m \left(1 + \frac{\|C\|_\infty}{\eta} \right) \right).$$

In particular, if the marginals have the same total mass, then $d_1 = \dots = d_m$, hence $\rho = 0$, and

$$\|u^*\|_\infty \leq \mathcal{R}, \quad |t^*| = \mathcal{O}(m\mathcal{R}).$$

Proof For

$$u = (u_1^\top, \dots, u_m^\top)^\top,$$

define

$$B_{i_1, \dots, i_m}(u) := \exp \left(\sum_{k=1}^m u_{k, i_k} - \frac{C_{i_1, \dots, i_m}}{\eta} \right)$$

and

$$S(u, t) := e^t \|B(u)\|_1 + \sum_{k=1}^m \|\exp(u_k)\|_1.$$

Recall that the dual objective is

$$\varphi(u, t) = D \log S(u, t) - \sum_{k=1}^m \langle u_k, p_k \rangle - st.$$

Let $(\hat{u}, \hat{t})$ be an optimal solution. Since the dual objective is smooth and unconstrained, the first-order optimality conditions give, for every $k \in [m]$ and $j \in [n]$,

$$p_{k,j} = \frac{D}{S(\hat{u}, \hat{t})} \left(e^{\hat{t}} [r_k(B(\hat{u}))]_j + e^{\hat{u}_{k,j}} \right), \quad (31)$$

$$s = \frac{De^{\hat{t}} \|B(\hat{u})\|_1}{S(\hat{u}, \hat{t})}. \quad (32)$$

For convenience, write

$$\alpha := \frac{D}{S(\hat{u}, \hat{t})}.$$

Thus,

$$p_{k,j} = \alpha \left(e^{\hat{t}} [r_k(B(\hat{u}))]_j + e^{\hat{u}_{k,j}} \right), \quad (33)$$

$$s = \alpha e^{\hat{t}} \|B(\hat{u})\|_1. \quad (34)$$

We first control the oscillation of each potential $\hat{u}_k$. Fix $k \in [m]$ and define

$$A_{k,j} := \sum_{i_{-k}} \exp \left(\sum_{\ell \neq k} \hat{u}_{\ell, i_{\ell}} - \frac{C_{i_1, \dots, i_{k-1}, j, i_{k+1}, \dots, i_m}}{\eta} \right), \quad j \in [n],$$

where i_{-k} denotes all indices except i_k . By construction,

$$[r_k(B(\hat{u}))]_j = e^{\hat{u}_{k,j}} A_{k,j},$$

and therefore (33) becomes

$$p_{k,j} = \alpha e^{\hat{u}_{k,j}} \left(1 + e^{\hat{t}} A_{k,j} \right). \quad (35)$$

Let $j, \ell \in [n]$. For every fixed choice of i_{-k} ,

$$-\|C\|_{\infty} \leq C_{i_1, \dots, j, \dots, i_m} - C_{i_1, \dots, \ell, \dots, i_m} \leq \|C\|_{\infty}.$$

Hence

$$e^{-\|C\|_{\infty}/\eta} \leq \frac{\exp(-C_{i_1, \dots, j, \dots, i_m}/\eta)}{\exp(-C_{i_1, \dots, \ell, \dots, i_m}/\eta)} \leq e^{\|C\|_{\infty}/\eta}.$$

The remaining factor

$$\exp \left(\sum_{r \neq k} \hat{u}_{r, i_r} \right)$$

is identical in the two sums. Since all summands are nonnegative, summing over i_{-k} preserves these multiplicative bounds:

$$e^{-\|C\|_{\infty}/\eta} A_{k,\ell} \leq A_{k,j} \leq e^{\|C\|_{\infty}/\eta} A_{k,\ell}. \quad (36)$$

If $a, b > 0$ satisfy $e^{-L}b \leq a \leq e^Lb$, then, for any $\lambda > 0$,

$$e^{-L} \leq \frac{1 + \lambda a}{1 + \lambda b} \leq e^L.$$

Indeed,

$$1 + \lambda a \leq 1 + \lambda e^L b \leq e^L (1 + \lambda b),$$

and the lower bound follows symmetrically. Applying this observation to (36) with

$$L = \frac{\|C\|_\infty}{\eta}, \quad \lambda = e^{\hat{t}},$$

gives

$$\left| \log \frac{1 + e^{\hat{t}} A_{k,j}}{1 + e^{\hat{t}} A_{k,\ell}} \right| \leq \frac{\|C\|_\infty}{\eta}. \quad (37)$$

Taking the ratio of (35) at j and ℓ , the normalization factor α cancels:

$$\frac{p_{k,j}}{p_{k,\ell}} = e^{\hat{u}_{k,j} - \hat{u}_{k,\ell}} \frac{1 + e^{\hat{t}} A_{k,j}}{1 + e^{\hat{t}} A_{k,\ell}}.$$

Consequently,

$$\hat{u}_{k,j} - \hat{u}_{k,\ell} = \log \frac{p_{k,j}}{p_{k,\ell}} - \log \frac{1 + e^{\hat{t}} A_{k,j}}{1 + e^{\hat{t}} A_{k,\ell}},$$

and (37) yields

$$|\hat{u}_{k,j} - \hat{u}_{k,\ell}| \leq \frac{\|C\|_\infty}{\eta} + \left| \log \frac{p_{k,j}}{p_{k,\ell}} \right|. \quad (38)$$

Let

$$p_{\min} := \min_{\substack{r \in [m] \\ q \in [n]}} p_{r,q}, \quad P_{\max} := \max_{r \in [m]} \|p_r\|_1.$$

Since

$$p_{\min} \leq p_{k,j}, p_{k,\ell} \leq \|p_k\|_1 \leq P_{\max},$$

both ratios $p_{k,j}/p_{k,\ell}$ and $p_{k,\ell}/p_{k,j}$ are at most $P_{\max}/p_{\min}$. Therefore,

$$\left| \log \frac{p_{k,j}}{p_{k,\ell}} \right| \leq \log \frac{P_{\max}}{p_{\min}}.$$

By the definition of $\mathcal{R}$,

$$|\hat{u}_{k,j} - \hat{u}_{k,\ell}| \leq \mathcal{R}.$$

Since j, ℓ are arbitrary,

$$\max_{j \in [n]} \hat{u}_{k,j} - \min_{j \in [n]} \hat{u}_{k,j} \leq \mathcal{R}, \quad k \in [m]. \quad (39)$$

We next choose a convenient representative of the optimal dual solution. In the classical multimarginal OT dual one may normalize the potential blocks separately. Here the slack terms couple these normalizations, and only a common shift remains. The relative locations of the blocks are therefore determined by the slack masses.

To see this, define

$$X_I := \alpha e^{\hat{t}} B_I(\hat{u}), \quad q_{k,j} := \alpha e^{\hat{u}_{k,j}}.$$

Equations (33)–(34) are precisely

$$r_k(X) + q_k = p_k, \quad \|X\|_1 = s.$$

Thus, summing the first identity over the k -th marginal,

$$\|q_k\|_1 = \|p_k\|_1 - s =: d_k.$$

Since $q_{k,j} = \alpha e^{\hat{u}_{k,j}}$,

$$d_k = \alpha \sum_{j=1}^n e^{\hat{u}_{k,j}}. \quad (40)$$

Define the log-mean-exp level of the k -th block by

$$a_k := \log \left(\frac{1}{n} \sum_{j=1}^n e^{\hat{u}_{k,j}} \right).$$

Then (40) gives

$$a_k = \log d_k - \log(\alpha n), \quad \text{and hence} \quad a_k - a_\ell = \log \frac{d_k}{d_\ell}. \quad (41)$$

Moreover,

$$\min_j \hat{u}_{k,j} \leq a_k \leq \max_j \hat{u}_{k,j}.$$

Together with (39), this implies

$$|\hat{u}_{k,j} - a_k| \leq \mathcal{R}. \quad (42)$$

It remains only to center the numbers a_k . The dual objective is invariant under the common shift

$$u_k \mapsto u_k + c \mathbf{1}_n, \quad t \mapsto t - (m-1)c, \quad c \in \mathbb{R}. \quad (43)$$

Indeed,

$$B(u_1 + c \mathbf{1}_n, \dots, u_m + c \mathbf{1}_n) = e^{mc} B(u),$$

so that

$$S(u_1 + c \mathbf{1}_n, \dots, u_m + c \mathbf{1}_n, t - (m-1)c) = e^c S(u, t).$$

Consequently, the log-partition term changes by Dc , whereas the linear terms change by

$$-c \sum_{k=1}^m \|p_k\|_1 + (m-1)sc.$$

These changes cancel because

$$D = \sum_{k=1}^m \|p_k\|_1 - (m-1)s.$$

Let

$$a_{\max} := \max_k a_k, \quad a_{\min} := \min_k a_k,$$

and choose

$$c := -\frac{a_{\max} + a_{\min}}{2}.$$

Denote the shifted optimal solution by (u^*, t^*) . Its block levels are $a_k^* = a_k + c$, and hence

$$\max_k a_k^* = -\min_k a_k^* = \frac{a_{\max} - a_{\min}}{2}.$$

By (41),

$$a_{\max} - a_{\min} = \log \frac{d_{\max}}{d_{\min}} =: \rho.$$

Therefore

$$|a_k^*| \leq \frac{\rho}{2}.$$

Since a common shift leaves $u_{k,j} - a_k$ unchanged, (42) gives

$$|u_{k,j}^*| \leq |u_{k,j}^* - a_k^*| + |a_k^*| \leq \mathcal{R} + \frac{\rho}{2}.$$

Thus

$$\|u^*\|_{\infty} \leq \mathcal{R} + \frac{\rho}{2}. \quad (44)$$

Notice that when all marginals have the same total mass, $d_1 = \dots = d_m$. In this case $\rho = 0$, all block levels a_k are equal, and the single common shift centers all potential blocks simultaneously. Hence (44) reduces simply to

$$\|u^*\|_{\infty} \leq \mathcal{R}.$$

We finally bound the scalar mass variable t^* . The role of the shift above is now useful: it has selected an optimal representative for which u^* is uniformly bounded, and the mass-balance equation then determines t^* up to quantities that can be bounded directly.

Let

$$\alpha^* := \frac{D}{S(u^*, t^*)}.$$

Applying the optimality conditions to the shifted solution gives

$$s = \alpha^* e^{t^*} \|B(u^*)\|_1.$$

On the other hand, summing the slack masses yields

$$D - s = \sum_{k=1}^m d_k = \alpha^* \sum_{k=1}^m \|\exp(u_k^*)\|_1.$$

Dividing the two identities eliminates the normalization α^* :

$$\frac{s}{D - s} = e^{t^*} \frac{\|B(u^*)\|_1}{\sum_{k=1}^m \|\exp(u_k^*)\|_1}. \quad (45)$$

Equivalently,

$$t^* = \log \frac{s}{D - s} + \log \left(\sum_{k=1}^m \|\exp(u_k^*)\|_1 \right) - \log \|B(u^*)\|_1. \quad (46)$$

Set

$$U := \mathcal{R} + \frac{\rho}{2}.$$

By (44),

$$-U \leq u_{k,j}^* \leq U \quad \text{for all } k, j.$$

Therefore,

$$ne^{-U} \leq \|\exp(u_k^*)\|_1 \leq ne^U,$$

and summing over k gives

$$mne^{-U} \leq \sum_{k=1}^m \|\exp(u_k^*)\|_1 \leq mne^U. \quad (47)$$

Similarly,

$$-mU \leq \sum_{k=1}^m u_{k,i_k}^* \leq mU.$$

Since $0 \leq C_I \leq \|C\|_\infty$,

$$e^{-mU - \|C\|_\infty/\eta} \leq B_I(u^*) \leq e^{mU}.$$

There are n^m entries in $B(u^*)$, so

$$n^m e^{-mU - \|C\|_\infty/\eta} \leq \|B(u^*)\|_1 \leq n^m e^{mU}. \quad (48)$$

Substituting (47) and (48) into (46), we obtain

$$\Gamma - (m+1)U \leq t^* \leq \Gamma + (m+1)U + \frac{\|C\|_\infty}{\eta},$$

where

$$\Gamma := \log \frac{s}{D-s} + \log m - (m-1) \log n.$$

Consequently,

$$|t^*| \leq |\Gamma| + (m+1) \left(\mathcal{R} + \frac{\rho}{2} \right) + \frac{\|C\|_\infty}{\eta}. \quad (49)$$

In particular, if all marginal total masses are equal to some M , then

$$d_k = M - s, \quad D - s = m(M - s), \quad \rho = 0,$$

and therefore

$$\Gamma = \log \frac{s}{M-s} - (m-1) \log n.$$

Moreover,

$$p_{\min} \leq \frac{M}{n} \implies \log n \leq \log \frac{M}{p_{\min}} \leq \mathcal{R},$$

and

$$\frac{\|C\|_\infty}{\eta} \leq \mathcal{R}.$$

Hence, for fixed transported-mass ratio $s/(M-s)$, (49) yields

$$|t^*| = \mathcal{O}(m\mathcal{R}).$$

This completes the proof. ■

3 Coordinate descent based algorithm

In this section, we focus on the design and analysis of algorithms that share a common foundation in *Sinkhorn-style* iterations, a class of methods that are well studied and widely implemented in the optimal transport (OT) literature. Such iterations can be interpreted as coordinate descent methods applied to the corresponding dual problem; however, this perspective alone does not fully account for the strong theoretical guarantees and the fast practical convergence observed for Sinkhorn-based algorithms. It was first observed in Altschuler et al. (2017) that, by leveraging Pinsker’s inequality to obtain sharper estimates on the objective gap decrease, Sinkhorn’s algorithm achieves a computational complexity of $\mathcal{O}\left(\frac{n^2}{\epsilon^3}\right)$. Subsequent works further exploited the shift-invariance structure of the smooth entropic OT dual to improve this bound to $\mathcal{O}\left(\frac{n^2}{\epsilon^2}\right)$. While the mathematical understanding of entropic OT is now relatively mature, analogous results for partial optimal transport (POT) and multi-marginal partial optimal transport (MMPOT) remain limited. In particular, Nguyen et al. (2024) shows that the standard approach of introducing dummy points to reduce POT to a classical OT problem, followed by Sinkhorn iterations, leads to a computational complexity of $\mathcal{O}\left(\frac{n^2}{\epsilon^4}\right)$. Moreover, for entropic MMPOT, existing coordinate descent methods are only known to converge asymptotically, and their iterative updates are typically non-differentiable, making it difficult to establish near-linear convergence guarantees. Motivated by these limitations, we aim to bridge this gap by proposing a Sinkhorn-style algorithm tailored to the MMPOT setting, which we refer to as GREENKHORNMMMPOT. The full algorithm description is given at Algorithm 1. The ρ function is defined as:

$$\rho(a, b) = \mathbf{1}_n^T(b - a) + \sum_{i=1}^n a_i \log \frac{a_i}{b_i}. \quad (50)$$

3.1 Convergence analysis

References

- Jason Altschuler, Jonathan Niles-Weed, and Philippe Rigollet. Near-linear time approximation algorithms for optimal transport via sinkhorn iteration. *Advances in neural information processing systems*, 30, 2017.
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- Anh Duc Nguyen, Tuan Dung Nguyen, Quang Minh Nguyen, Hoang H Nguyen, Lam M Nguyen, and Kim-Chuan Toh. On partial optimal transport: Revising the infeasibility of sinkhorn and efficient gradient methods. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 8090–8098, 2024.

Algorithm 1 GreenkhornMMPOT

Require: Cost tensor $C \in \mathbb{R}^{n \times \dots \times n}$, regularization parameter $\eta > 0$, marginals $p_1, \dots, p_m$, total mass s , and tolerance $\epsilon > 0$.

```

1: while  $\text{dist}(X^{(\tau)}, p, s) > \epsilon$  do
2:   // Identify the constraint with the maximum current error
3:    $I \leftarrow \arg \max_{k \in [m]} \rho \left( p_k, r_k(X^{(\tau)}) + q_k^{(\tau)} \right)$ 
4:   if  $\rho \left( p_I, r_I(X^{(\tau)}) + q_I^{(\tau)} \right) > \rho(s, S(X^{(\tau)}))$  then
5:      $u_I^{(\tau+1)} \leftarrow u_I^{(\tau)} + \log(p_I) - \log \left( r_I(X^{(\tau)}) + q_I^{(\tau)} \right)$ 
6:      $u_k^{(\tau+1)} \leftarrow u_k^{(\tau)}, \quad \forall k \neq I$ 
7:      $t^{(\tau+1)} \leftarrow t^{(\tau)}$ 
8:   else
9:      $t^{(\tau+1)} \leftarrow t^{(\tau)} + \log(s) - \log(S(X^{(\tau)}))$ 
10:     $u_k^{(\tau+1)} \leftarrow u_k^{(\tau)}, \quad \forall k \in [m]$ 
11:   end if
12:   Update  $X^{(\tau+1)}$  and  $q_k^{(\tau+1)}$  via the new dual variables  $u^{(\tau+1)}$  and  $t^{(\tau+1)}$ .
13:    $\tau \leftarrow \tau + 1$ 
14: end while
15: return  $u^{(\tau)}, t^{(\tau)}, X^{(\tau)}$ 

```
